CUET PG Computer Science / MCA Entrance Exam

CUET PG 200+ High-Yield Question Bank

Digital Fundamentals & Computer Logic Architecture

* Comprehensive coverage of all 17 lectures & core Digital Fundamentals units
* Explained in beginner-friendly Hinglish with step-by-step logic and intuition
* Complete logic gate circuit diagrams, truth tables, and K-Map simplifications
* Meticulously solved step-by-step calculations, formula sheets & exam traps
* Designed specifically for CUET PG, MCA, and State Entrance Examinations

[1] Master Conceptual Notes

All 17 Units: Number Systems, Complements, Floating Point, Boolean Algebra, Gates, K-Maps, Adders/ Subtractors, MUX, Latches, Flip-Flops, Registers, Counters & Memory Hierarchy.

[2] Exam-Oriented Focus

Detailed error traps, quick revision formulas, difference tables, step-by-step solved numericals, and CUET PG expected questions with complete Hinglish explanations.

Target: CUET PG Computer Science / MCA Entrance Exams

Material Type: Comprehensive Study Notes & Problem Workbook
Syllabus Coverage: Complete 17 Units (Lectures 01 to 17)
Language: Easy Hinglish (Bilingual Hindi + English explanation for rapid conceptual mastery)
Edition: Updated Exam Edition

EXAM INSTRUCTIONS & TOPIC WEIGHTAGE

[MUST REMEMBER FOR CUET PG] CUET PG MCA / CS Marking Scheme & Scoring Rules:
- Correct Answer: +4 Marks
- Incorrect Answer (Negative Marking): -1 Mark (25% penalty)
- Unattempted Question: 0 Marks
- Exam Strategy: Pehle pass mein direct factual aur formula-based questions attempt karein (under 40 seconds). Numerical conversions (binary/hex) aur K-Map questions ko second pass mein solve karein. Jab

CUET PG Marking Scheme & Examination Strategy

Syllabus Breakdown of 225 Practice Questions across all 17 Units:

Unit / Module	Topics Covered	No. of MCQs	Exam Priority
Units 1 to 4	Data Units, Base Conversions, 1s/2s Comp, IEEE 754 Float	62 Questions	Very High (15-20%)
Units 5 to 6	BCD, Excess-3, Gray Code, Booth Algorithm, CPU Buses	25 Questions	High (10-12%)
Units 7 to 11	Boolean Laws, Universal Gates, SOP/POS, 2-4 Var K-Maps	63 Questions	Very High (20-25%)
Units 12 to 13	Half/Full Adders, Subtractors, MUX, DEMUX, Decoders	30 Questions	High (12-15%)
Units 14 to 16	Latches, SR/JK/D/T Flip-Flops, Shift Registers, Counters	37 Questions	High (12-15%)
Unit 17	Memory Organization, SRAM vs DRAM, Cache Mapping, AMAT	8 Questions	Medium-High (6-8%)
Grand Total	Complete 17 Units of Digital Fundamentals Syllabus	225 Questions	100% Coverage

[EXAM TRAP / COMMON CONFUSION] Top 3 Common Traps in CUET PG Digital Fundamentals:
1. Complement Ranges: Unsigned n-bit range is 0 to 2^n - 1, whereas 2's complement is -2^(n-1) to +2^(n-1) - 1. Exam mein signed vs unsigned dhyan se padhein!
2. Floating Point Bias: IEEE 754 Single Precision uses Bias = 127 (NOT 128). Actual exponent = E - 127.
3. Universal Gate Count: Minimum NAND gates for XOR is 4, for XNOR is 5. Minimum NOR gates for XOR is

Common High-Risk Traps in CUET PG

UNIT 1: DATA UNITS & INTRO - PRACTICE MCQS

Q1. Which of the following is the smallest addressable unit of data in modern computer architecture? Easy

(A) Bit
(B) Byte
(C) Nibble
(D) Word

Correct Answer: [B]
Explanation: Bit data ki absolute smallest binary unit hoti hai (0 ya 1), lekin modern computer architectures mein memory ko address karne ki smallest unit Byte (8 bits) hoti hai.

Q2. A group of 4 consecutive binary bits is known as a: Easy

(A) Byte
(B) Octet
(C) Nibble
(D) Word

Correct Answer: [C]
Explanation: 4 bits ke group ko 1 Nibble kehte hain. 1 Byte mein 2 Nibbles hote hain. Ek single hexadecimal digit 1 nibble ko represent karta hai.

Q3. How many bits are present in 1 Kibibyte (KiB) according to binary IEC standards? Medium

(A) 1,000 bits
(B) 8,000 bits
(C) 1,024 bits
(D) 8,192 bits

Correct Answer: [D]
Explanation: 1 KiB = 1,024 Bytes. 1 Byte = 8 bits. Isliye total bits = 1024 * 8 = 8,192 bits.

Q4. Arrange the following memory units in increasing order of size: (A) Terabyte, (B) Megabyte, (C) Exabyte, (D) Gigabyte, (E) Petabyte. Easy

(A) B < D < A < E < C
(B) B < A < D < E < C
(C) D < B < A < E < C
(D) A < B < D < E < C

Correct Answer: [A]
Explanation: Correct order of binary power units: Megabyte (2^20) < Gigabyte (2^30) < Terabyte (2^40) < Petabyte (2^50) < Exabyte (2^60). Thus B < D < A < E < C.

Q5. A digital signal is characterized by being: Easy

(A) Continuous in time and value
(B) Discrete in time and continuous in value
(C) Discrete in both time and amplitude values
(D) Dependent on ambient temperature

Correct Answer: [C]
Explanation: Digital signals discrete in time aur discrete in amplitude (voltage levels, e.g. 0V and 5V) hote hain, jabki analog signals continuous hote hain.

Q6. How many distinct bit patterns can be formed using 8 bits? Easy

(A) 64
(B) 128
(C) 256
(D) 512

Correct Answer: [C]
Explanation: n bits se total 2^n unique combinations bante hain. For n = 8 bits: 2^8 = 256 unique patterns (0 to 255).

Q7. If a network link has a transfer rate of 64 Megabits per second (Mbps), its speed in Megabytes per second (MBps) is: Easy

(A) 8 MBps
(B) 16 MBps
(C) 32 MBps
(D) 512 MBps

Correct Answer: [A]
Explanation: 1 Byte = 8 bits. 64 Megabits / 8 = 8 Megabytes per second (8 MBps).

Q8. Which of the following is a major advantage of digital systems over analog systems? Easy

(A) Infinite resolution
(B) High noise immunity and easy error correction
(C) Low circuit complexity for natural sound recording
(D) Direct compatibility with biological organs

Correct Answer: [B]
Explanation: Digital circuits noise immune hote hain kyunki thoda bohot voltage fluctuation 0 ya 1 ke threshold logic ko affect nahi karta.

Q9. In a 64-bit computer architecture, the term "64-bit" specifically refers to: Medium

(A) Total hard disk capacity
(B) Size of the CPU word / internal registers
(C) Clock speed of the CPU
(D) Number of keys on the keyboard

Correct Answer: [B]
Explanation: Processor architecture word size (size of general purpose registers, ALU width, aur address bus capability) ko define karti hai.

Q10. How many nibbles are present in 16 Kilobytes (KB)? Medium

(A) 16,384
(B) 32,768
(C) 65,536
(D) 8,192

Correct Answer: [B]
Explanation: 1 Byte = 2 Nibbles. 16 KB = 16 * 1024 Bytes = 16,384 Bytes. Nibbles = 16,384 * 2 = 32,768 nibbles.

Q11. In digital logic, active-low signal means: Easy

(A) Signal is active when voltage is High (Logic 1)
(B) Signal is active when voltage is Low (Logic 0)
(C) Signal cannot carry data
(D) Signal voltage is negative

Correct Answer: [B]
Explanation: Active-low signals tab assert/trigger hote hain jab unki value Logic 0 (Low) hoti hai (usually indicated by a bar like CS' or EN').

Q12. What power of 2 represents 1 Petabyte (PB)? Easy

(A) 2^30 Bytes
(B) 2^40 Bytes
(C) 2^50 Bytes
(D) 2^60 Bytes

Correct Answer: [C]
Explanation: KB = 2^10, MB = 2^20, GB = 2^30, TB = 2^40, PB = 2^50 Bytes.

UNIT 2: NUMBER SYSTEMS - PRACTICE MCQS

Q13. The decimal equivalent of the binary number (110101)_2 is: Easy

(A) 51
(B) 53
(C) 55
(D) 57

Correct Answer: [B]
Explanation: 1*32 + 1*16 + 0*8 + 1*4 + 0*2 + 1*1 = 32 + 16 + 4 + 1 = 53.

Q14. Convert the decimal number (75)_10 to its Octal equivalent: Easy

(A) 113_8
(B) 111_8
(C) 115_8
(D) 123_8

Correct Answer: [A]
Explanation: 75 / 8 = 9 rem 3; 9 / 8 = 1 rem 1; 1 / 8 = 0 rem 1. Reading bottom to top gives (113)_8.

Q15. Convert the binary number (101111011100)_2 directly to Hexadecimal: Medium

(A) BDC_16
(B) ADC_16
(C) BCC_16
(D) CDB_16

Correct Answer: [A]
Explanation: 4-bit grouping from right: [1011] [1101] [1100]. 1011 = B (11), 1101 = D (13), 1100 = C (12). So BDC_16.

Q16. The fractional binary number (0.1011)_2 in decimal is equal to: Medium

(A) 0.6875 (B) 0.6250
(C) 0.7500 (D) 0.8125

Correct Answer: [A]
Explanation: 1*2^-1 + 0*2^-2 + 1*2^-3 + 1*2^-4 = 0.5 + 0 + 0.125 + 0.0625 = 0.6875.

Q17. Convert Hexadecimal (2F)_16 to decimal: Easy

(A) 45 (B) 47
(C) 49 (D) 51

Correct Answer: [B]
Explanation: 2 * 16^1 + F * 16^0 = 2 * 16 + 15 * 1 = 32 + 15 = 47.

Q18. Convert Octal (353)_8 into binary: Easy

(A) 011101011 (B) 011110011
(C) 101011011 (D) 111001011

Correct Answer: [A]
Explanation: Har octal digit ko 3-bit binary mein likho: 3 -> 011, 5 -> 101, 3 -> 011. Result = 011101011.

Q19. What is the sum of hexadecimal numbers (7CA)_16 + (4B2)_16? Medium

(A) B7C_16 (B) C7C_16
(C) D7C_16 (D) B6C_16

Correct Answer: [B]
Explanation: A (10) + 2 = 12 = C. C (12) + B (11) = 23 = 16 + 7 (7 with carry 1). 7 + 4 + 1 = 12 = C. Result = C7C_16.

Q20. If (121)_r = (100)_8, what is the value of base r? Hard

(A) 6 (B) 7
(C) 8 (D) 9

Correct Answer: [B]
Explanation: (100)_8 = 1*8^2 + 0 + 0 = 64_10. (121)_r = 1*r^2 + 2*r + 1 = (r+1)^2. So (r+1)^2 = 64 => r+1 = 8 => r = 7.

Q21. The number of 1s in the binary representation of 3*4096 + 15*256 + 5*16 + 3 is: Hard

(A) 8 (B) 9
(C) 10 (D) 12

Correct Answer: [C]
Explanation: Notice powers of 16: 4096 = 16^3, 256 = 16^2, 16 = 16^1, 1 = 16^0. In Hex: (3 F 5 3)_16. Now write each hex digit in 4-bit binary: 3 -> 0011 (two 1s), F -> 1111 (four 1s), 5 -> 0101 (two 1s), 3 -> 0011 (two 1s). Total 1s = 2 + 4 + 2 + 2 = 10!

Q22. Convert decimal (10.25)_10 into binary: Easy

(A) 1010.10 (B) 1010.01
(C) 1010.00 (D) 1010.11

Correct Answer: [B]
Explanation: 10 in binary = 1010. 0.25 * 2 = 0.50 (int 0); 0.50 * 2 = 1.00 (int 1). So (10.25)_10 = (1010.01)_2.

Q23. How many binary bits are required to represent an octal number of 4 digits? Easy

(A) 8 bits
(B) 12 bits
(C) 16 bits
(D) 4 bits

Correct Answer: [B]
Explanation: Har octal digit 3 bits leta hai. 4 octal digits = 4 * 3 = 12 binary bits.

Q24. Convert (101010011.110100)_2 to octal: Medium

(A) 523.64_8
(B) 533.64_8
(C) 543.65_8
(D) 533.56_8

Correct Answer: [A]
Explanation: 3-bit grouping: [101] [010] [011] . [110] [100] = 5 2 3 . 6 4 => 523.64_8.

Q25. The radix of a number system having digits {0, 1, 2, 3, 4} is: Easy

(A) 4
(B) 5
(C) 6
(D) 10

Correct Answer: [B]
Explanation: A base-r system contains digits from 0 to r-1. Agar digits {0, 1, 2, 3, 4} hain to base r = 5 hai.

Q26. Which of the following numbers is an invalid octal number? Easy

(A) 457_8
(B) 702_8
(C) 381_8
(D) 100_8

Correct Answer: [C]
Explanation: Octal system mein sirf digits 0 to 7 allowed hoti hain. Digit '8' octal mein invalid hai!

Q27. The hexadecimal equivalent of octal number (157)_8 is: Medium

(A) 6F_16
(B) 7F_16
(C) 8F_16
(D) 5F_16

Correct Answer: [A]
Explanation: (157)_8 to binary: [001] [101] [111] = 001101111. Regroup in 4 bits: [0110] [1111] = 6 F => (6F)_16.

Q28. Evaluate: (1101)_2 * (101)_2 in binary: Medium

(A) (1000001)_2
(B) (100001)_2
(C) (1000011)_2
(D) (111101)_2

Correct Answer: [A]
Explanation: (1101)_2 = 13_10, (101)_2 = 5_10. Product = 13 * 5 = 65_10. 65 in binary = 64 + 1 = 1000001_2.

Q29. If (24)_x = (14)_10, what is the value of base x? Medium

(A) 5
(B) 6
(C) 7
(D) 8

Correct Answer: [A]
Explanation: 2*x + 4 = 14 => 2x = 10 => x = 5.

Q30. Convert binary fraction (0.011)_2 to decimal: Easy

(A) 0.375 (B) 0.25
(C) 0.125 (D) 0.625

Correct Answer: [A]
Explanation: 0*2^-1 + 1*2^-2 + 1*2^-3 = 0.25 + 0.125 = 0.375.

Q31. The decimal value of (211)_8 is: Easy

(A) 135 (B) 137
(C) 139 (D) 141

Correct Answer: [B]
Explanation: 2 * 8^2 + 1 * 8^1 + 1 * 8^0 = 2 * 64 + 8 + 1 = 128 + 9 = 137.

Q32. How many hex digits are required to represent a 32-bit binary memory address? Easy

(A) 4 (B) 6
(C) 8 (D) 16

Correct Answer: [C]
Explanation: 1 Hex digit = 4 bits. 32 bits / 4 = 8 Hexadecimal digits.

UNIT 3: COMPLEMENTS & FIXED POINT - PRACTICE MCQS

Q33. The 1's complement of binary number 1011001 is: Easy

(A) 0100110 (B) 0100111
(C) 1011010 (D) 0100100

Correct Answer: [A]
Explanation: Saare bits invert kar do: 1->0, 0->1. 1011001 becomes 0100110.

Q34. The 2's complement of binary number 1011000 is: Easy

(A) 0100111 (B) 0101000
(C) 0100100 (D) 1011001

Correct Answer: [B]
Explanation: Right se pehle 1 tak same (...1000), baaki left bits invert: 101 becomes 010. Result = 0101000.

Q35. What is the range of integers that can be represented using 8-bit 2's complement form? Easy

(A) -128 to +127 (B) -127 to +128
(C) -127 to +127 (D) -256 to +255

Correct Answer: [A]
Explanation: Range of n-bit 2's complement is -2^(n-1) to +(2^(n-1) - 1). For n=8: -128 to +127.

Q36. In 8-bit 2's complement representation, the bit pattern 10000000 represents: Medium

(A) +0 (B) -0
(C) -128 (D) +128

Correct Answer: [C]
Explanation: 10000000 represents the most negative number: -2^(8-1) = -128.

Q37. Representation of decimal -11 in 5-bit Signed Magnitude is: Medium

(A) 11011
(B) 01011
(C) 10101
(D) 11101

Correct Answer: [A]
Explanation: 5-bit format: MSB is sign bit (1 for negative). Magnitude 11 in 4 bits = 1011. Combine: 1 1011 = 11011.

Q38. Statement I: 011010 is 2's complement representation of -37.
Statement II: (211)_8 is octal representation of 137.
Choose the correct option: Hard

(A) Both are true
(B) Both are false
(C) Statement I is true, II is false
(D) Statement I is false, II is true

Correct Answer: [D]
Explanation: Statement I: 011010 has MSB 0 (positive number +26), cannot represent -37. False. Statement II: 2*64 + 1*8 + 1 = 137. True. Correct option is D.

Q39. When two 4-bit 2's complement numbers 1101 (-3) and 0100 (+4) are added, the result is: Medium

(A) 0001 with overflow
(B) 1001 with no overflow
(C) 1001 with overflow
(D) 0001 with no overflow

Correct Answer: [D]
Explanation: -3 + 4 = +1. In 4-bit 2's complement, +1 = 0001. Ek negative aur ek positive number ko add karne par overflow KABHI NAHI ho sakta. So 0001 and no overflow.

Q40. What is the 9's complement of the decimal number 456? Easy

(A) 543
(B) 544
(C) 554
(D) 643

Correct Answer: [A]
Explanation: 999 - 456 = 543.

Q41. What is the 10's complement of the decimal number 456? Easy

(A) 543
(B) 544
(C) 545
(D) 644

Correct Answer: [B]
Explanation: 10's complement = 9's complement + 1 = 543 + 1 = 544.

Q42. Arithmetic overflow during addition of two signed 2's complement numbers occurs when: Medium

(A) Carry into MSB equals carry out of MSB
(B) Carry into MSB is different from carry out of MSB (Cin XOR Cout = 1)
(C) Both operands have different signs
(D) The sum is zero

Correct Answer: [B]
Explanation: Overflow condition in 2's complement: V = C_in XOR C_out = 1. Jab do same-sign numbers ka sum opposite sign ka ban jaye.

Q43. How many representations of ZERO exist in 1's complement and 2's complement respectively for n bits? Easy

(A) 1 and 2
(B) 2 and 1
(C) 2 and 2
(D) 1 and 1

Correct Answer: [B]
Explanation: 1's complement mein do zeros hote hain (+0 = 0000, -0 = 1111). 2's complement mein sirf ek unique zero hota hai (0000).

Q44. The decimal equivalent of 8-bit 2's complement number 11111110 is: Easy

(A) -1
(B) -2
(C) -3
(D) -254

Correct Answer: [B]
Explanation: MSB is 1, so negative. Magnitude = 2's complement of 11111110 = 00000001 + 1 = 00000010 = 2. So value = -2.

Q45. Which system is preferred in arithmetic logic units (ALUs) for representing negative integers? Easy

(A) Signed Magnitude
(B) 1's Complement
(C) 2's Complement
(D) Biased representation

Correct Answer: [C]
Explanation: 2's complement ALU hardware design ko simple karta hai kyunki subtraction ko direct addition se implement kiya ja sakta hai aur zero unique hota hai.

Q46. In 2's complement subtraction A - B, if an end-around carry is generated at the MSB, we must: Medium

(A) Add it back to the LSB
(B) Discard / drop the carry
(C) Subtract it from the result
(D) Invert all bits

Correct Answer: [B]
Explanation: 2's complement mein carry-out ko DISCARD kar dete hain! (1's complement mein end-around carry ko add karte hain).

Q47. What is the 16-bit 2's complement representation of -1? Easy

(A) 1000 0000 0000 0001
(B) 1111 1111 1111 1111
(C) 0111 1111 1111 1111
(D) 1111 1111 1111 1110

Correct Answer: [B]
Explanation: +1 = 000...0001. 1's comp = 111...1110. Add 1 = 1111 1111 1111 1111. All 1s represents -1.

Q48. Sign extension of an 8-bit 2's complement number 11010000 to 16 bits results in: Medium

(A) 00000000 11010000
(B) 11111111 11010000
(C) 10000000 11010000
(D) 11010000 00000000

Correct Answer: [B]
Explanation: Sign extension mein MSB bit (sign bit) ko left side copy kiya jata hai. Yahan MSB 1 hai, isliye 8 ones prepend honge: 11111111 11010000.

Q49. What is the 7's complement of the octal number (562)_8? Medium

(A) 215_8
(B) 216_8
(C) 315_8
(D) 217_8

Correct Answer: [A]
Explanation: 777_8 - 562_8 = 215_8.

Q50. What is the 8's complement of the octal number (562)_8? Medium

(A) 215_8 (B) 216_8
(C) 217_8 (D) 316_8

Correct Answer: [B]
Explanation: 8's complement = 7's complement + 1 = 215_8 + 1 = 216_8.

UNIT 4: FLOATING POINT - PRACTICE MCQS

Q51. In IEEE 754 single precision (32-bit) format, the number of bits allocated to Sign, Exponent, and Mantissa are: Easy

(A) 1, 7, 24 (B) 1, 8, 23
(C) 1, 11, 52 (D) 1, 8, 24

Correct Answer: [B]
Explanation: 32-bit Single Precision: Sign = 1 bit, Biased Exponent = 8 bits, Mantissa/Fraction = 23 bits.

Q52. What is the bias value added to the true exponent in IEEE 754 single-precision float? Easy

(A) 128 (B) 127
(C) 255 (D) 1023

Correct Answer: [B]
Explanation: Single precision exponent has 8 bits, so bias = 2^(8-1) - 1 = 127.

Q53. In IEEE 754 double precision (64-bit) floating point standard, the exponent bias is: Easy

(A) 127 (B) 511
(C) 1023 (D) 2047

Correct Answer: [C]
Explanation: Double precision exponent has 11 bits, bias = 2^(11-1) - 1 = 1023.

Q54. In IEEE 754 floating point format, if the exponent field contains all 1s and the mantissa is all 0s, the value represents: Medium

(A) Zero (B) Denormalized number
(C) Infinity (+inf or -inf) (D) NaN (Not a Number)

Correct Answer: [C]
Explanation: E = All 1s, F = 0 represents Infinity. S=0 gives +inf, S=1 gives -inf.

Q55. If Exponent is all 1s and Mantissa is NON-ZERO in IEEE 754, it represents: Medium

(A) Zero (B) Infinity
(C) Underflow (D) NaN (Not a Number)

Correct Answer: [D]
Explanation: E = All 1s, Mantissa != 0 indicates NaN (Not a Number, jaise 0/0 ka result).

Q56. Why is an exponent bias used in floating point representation? Medium

(A) To allow negative exponents to be stored as positive unsigned integers
(B) To increase memory speed
(C) To eliminate the sign bit
(D) To prevent rounding errors

Correct Answer: [A]
Explanation: Bias add karne se actual positive aur negative exponents unsigned integer range (1 to 254) mein map ho jate hain, jisse hardware comparison simple ho jata hai.

Q57. Convert the decimal number -0.75 into IEEE 754 32-bit single precision floating point representation: Hard

(A) 1 01111110 10000000000000000000000
(B) 1 01111111 10000000000000000000000
(C) 0 01111110 10000000000000000000000
(D) 1 10000000 10000000000000000000000

Correct Answer: [A]
Explanation: 1. Negative => S = 1. 2. 0.75 = (0.11)_2 = 1.1 * 2^-1. True exp = -1. 3. Biased exp = -1 + 127 = 126 = 01111110. 4. Normalized fraction = 1 (followed by twenty-two 0s). Combined: 1 01111110 10000...0.

Q58. A normalized floating-point number has an implicit (hidden) leading bit of: Easy

(A) 0
(B) 1
(C) Depends on sign bit
(D) Does not exist

Correct Answer: [B]
Explanation: IEEE 754 normalized form mein radix point se pehle hamesha 1 assume kiya jata hai (1.M), jisse 1 bit memory save hoti hai.

Q59. In IEEE 754 standard, zero is represented by: Medium

(A) Exponent all 0s, Fraction all 0s
(B) Exponent all 1s, Fraction all 0s
(C) All 32 bits as 1
(D) Exponent = 127, Fraction = 0

Correct Answer: [A]
Explanation: E = 00000000 and F = 000...00 represents Zero (S=0 gives +0, S=1 gives -0).

Q60. Denormalized numbers in IEEE 754 format allow for: Hard

(A) Faster multiplication
(B) Graceful underflow near zero
(C) Handling infinite recursion
(D) Eliminating cache miss

Correct Answer: [B]
Explanation: Denormalized numbers (E = 0, F != 0, implicit leading bit 0) zero ke pass ke bohot chhote numbers ko represent karne aur graceful underflow ke liye use hote hain.

Q61. If a 12-bit float has 1 sign bit, 4 exponent bits (bias 7), and 7 fraction bits, what is +infinity? Medium

(A) 0 1111 0000000
(B) 1 1111 0000000
(C) 0 0000 1111111
(D) 0 1111 1111111

Correct Answer: [A]
Explanation: +inf => Sign = 0, Exponent = All 1s (1111), Fraction = All 0s (0000000).

Q62. What is the true exponent value if the stored 8-bit biased exponent is (10000010)_2 in single precision? Medium

(A) +2
(B) +3
(C) +4
(D) +130

Correct Answer: [B]
Explanation: (10000010)_2 = 130_10. True exponent = Biased exponent - 127 = 130 - 127 = +3.

UNIT 5: COMPUTER ARITHMETIC & CODES - PRACTICE MCQS

Q63. Booth's algorithm is primarily used for: Easy

(A) Binary division of floating point numbers
(B) Fast multiplication of signed 2's complement numbers
(C) Converting Gray code to binary
(D) Error detection in transmission

Correct Answer: [B]
Explanation: Booth's multiplication algorithm signed 2's complement binary integers ko efficiently multiply karne ke liye use hota hai.

Q64. In Booth's algorithm, when the multiplier bit pair (Q_i, Q_{i-1}) is (1, 0), the action taken is: Medium

(A) Add multiplicand to accumulator and shift
(B) Subtract multiplicand from accumulator and shift
(C) Only arithmetic shift right
(D) No operation

Correct Answer: [B]
Explanation: 1 0 indicates the beginning of a block of 1s in negative direction, so subtract multiplicand M (A = A - M), then ASR.

Q65. Which of the following is a self-complementing digital code? Medium

(A) BCD (8421)
(B) Excess-3 Code
(C) Gray Code
(D) ASCII Code

Correct Answer: [B]
Explanation: Excess-3 code self-complementing code hai: kisi digit ke Excess-3 code ka 1's complement lene par uske 9's complement ka Excess-3 code mil jata hai.

Q66. The Gray code equivalent of binary number 1011 is: Medium

(A) 1110
(B) 1101
(C) 1010
(D) 1111

Correct Answer: [A]
Explanation: G3 = B3 = 1; G2 = 1 XOR 0 = 1; G1 = 0 XOR 1 = 1; G0 = 1 XOR 1 = 0. Result = 1110.

Q67. Convert Gray code 1101 to binary: Medium

(A) 1001
(B) 1010
(C) 1011
(D) 1111

Correct Answer: [A]
Explanation: B3 = G3 = 1; B2 = 1 XOR 1 = 0; B1 = 0 XOR 0 = 0; B0 = 0 XOR 1 = 1. Result = 1001.

Q68. Gray code is also known as: Easy

(A) Weighted code
(B) Unit-distance code / Cyclic code
(C) Alphanumeric code
(D) Error correcting code

Correct Answer: [B]
Explanation: Gray code mein do consecutive code words mein sirf ek bit change hoti hai, isliye ise Unit-Distance Code ya Cyclic Code kehte hain.

Q69. Which of the following 4-bit codes represents an INVALID BCD state? Easy

(A) 0111
(B) 1000
(C) 1001
(D) 1011

Correct Answer: [D]
Explanation: Valid BCD states 0000 to 1001 (0 to 9) hoti hain. 1010 se 1111 (10 to 15) invalid hoti hain. 1011 (11) invalid BCD hai.

Q70. When adding two BCD numbers, if the sum exceeds 9 or produces a carry, the correction step is: Medium

(A) Subtract 6 (0110)
(B) Add 6 (0110)
(C) Add 3 (0011)
(D) Subtract 3 (0011)

Correct Answer: [B]
Explanation: BCD addition mein agar sum > 9 ho ya carry out generate ho, to 6 (0110) add kiya jata hai 6 invalid states ko skip karne ke liye.

Q71. The decimal digit 7 represented in Excess-3 code is: Easy

(A) 0111
(B) 1000
(C) 1001
(D) 1010

Correct Answer: [D]
Explanation: Excess-3 = Decimal + 3 = 7 + 3 = 10 = 1010.

Q72. Which of the following is a non-weighted code? Easy

(A) BCD 8421
(B) Excess-3 Code
(C) Binary
(D) Octal

Correct Answer: [B]
Explanation: Excess-3 aur Gray code non-weighted hote hain (unke bits ki fixed positional weights jaise 8, 4, 2, 1 nahi hoti).

Q73. In Booth's algorithm, the worst-case number of addition/subtraction operations occurs for multiplier bit pattern: Hard

(A) 00000000
(B) 11111111
(C) 01010101
(D) 00001111

Correct Answer: [C]
Explanation: Alternating bits (01010101) mein har step par transition (01 ya 10) hota hai, jisse maximum operations execute hote hain.

Q74. Parity bits are added to digital data primarily for: Easy

(A) Data compression
(B) Single-bit error detection
(C) Data encryption
(D) Increasing clock frequency

Correct Answer: [B]
Explanation: Parity bit transmission ke dauran single-bit errors ko detect karne ke liye use hoti hai.

Q75. If even parity is used, what is the parity bit for the data word 1011010? Medium

(A) 0
(B) 1
(C) Either 0 or 1
(D) None

Correct Answer: [A]
Explanation: Data word 1011010 mein total char 1s hain (even number of 1s). Even parity maintain rakhne ke liye parity bit 0 honi chahiye.

UNIT 6: COMPUTER ARCHITECTURE - PRACTICE MCQS

Q76. Which CPU register holds the memory address of the NEXT instruction to be fetched and executed? Easy

(A) Instruction Register (IR)
(B) Program Counter (PC)
(C) Memory Address Register (MAR)
(D) Accumulator (AC)

Correct Answer: [B]
Explanation: Program Counter (PC) next instruction ka address hold karta hai. Instruction fetch hone ke baad PC automatically agle address par increment ho jata hai.

Q77. The Instruction Register (IR) holds: Easy

(A) Address of next instruction
(B) The instruction currently being executed / decoded
(C) Intermediate ALU result
(D) The interrupt vector

Correct Answer: [B]
Explanation: Instruction Register (IR) memory se fetch kiye gaye current instruction (opcode + operands) ko decode aur execute hone tak hold karta hai.

Q78. Which of the following registers acts as a direct buffer between the CPU internal bus and the system data bus? Easy

(A) Memory Address Register (MAR)
(B) Memory Data Register (MDR) / MBR
(C) Program Counter (PC)
(D) Flag Register

Correct Answer: [B]
Explanation: MDR / MBR data bus se directly connected hota hai aur CPU tatha Memory/IO devices ke beech data transfer buffer ka kaam karta hai.

Q79. The correct sequence of phases in a basic CPU instruction execution cycle is: Easy

(A) Execute -> Decode -> Fetch
(B) Decode -> Fetch -> Execute
(C) Fetch -> Decode -> Execute
(D) Fetch -> Execute -> Decode

Correct Answer: [C]
Explanation: Standard instruction cycle: 1. Fetch (memory se instruction lana), 2. Decode (opcode interpret karna), 3. Execute (operation perform karna).

Q80. Which type of Control Unit uses microcode stored in a control ROM? Medium

(A) Hardwired Control Unit
(B) Microprogrammed Control Unit
(C) Pipelined Control Unit
(D) Asynchronous Control Unit

Correct Answer: [B]
Explanation: Microprogrammed Control Unit control memory (ROM) mein stored microinstructions execute karke control signals generate karta hai.

Q81. Compared to a Microprogrammed Control Unit, a Hardwired Control Unit is: Medium

(A) Slower and easier to modify
(B) Faster and harder to modify
(C) Slower and harder to modify
(D) Identical in speed and flexibility

Correct Answer: [B]
Explanation: Hardwired CU pure combinational gates aur flip-flops se bana hota hai, isliye ye bohot fast hota hai par design badalna lagbhag impossible hota hai.

Q82. If a computer has a 24-bit address bus, what is the maximum directly addressable memory capacity (byte-addressable)? Medium

(A) 16 Megabytes (MB)
(B) 24 Megabytes (MB)
(C) 8 Megabytes (MB)
(D) 64 Megabytes (MB)

Correct Answer: [A]
Explanation: 2^24 Bytes = 2^4 * 2^20 Bytes = 16 * 1 MB = 16 MB.

Q83. Which system bus is strictly UNIDIRECTIONAL in a standard computer system? Medium

(A) Data Bus
(B) Control Bus
(C) Address Bus
(D) System Bus

Correct Answer: [C]
Explanation: Address Bus unidirectional hoti hai: address hamesha CPU (master) se memory ya I/O devices ki taraf jata hai.

Q84. The Accumulator register in a CPU is typically used to: Easy

(A) Store instructions
(B) Hold the address of memory
(C) Hold the initial operand and the result of ALU operations
(D) Generate clock pulses

Correct Answer: [C]
Explanation: Accumulator (AC) ALU ka primary operational register hota hai jisme calculations ke operands aur results store hote hain.

Q85. What is the function of the Memory Address Register (MAR)? Easy

(A) Holds the data read from memory
(B) Holds the physical address of the memory location being read or written
(C) Holds the instruction opcode
(D) Counts clock ticks

Correct Answer: [B]
Explanation: MAR address bus se directly connected hota hai aur targeted memory cell ka address hold karta hai.

Q86. During an instruction fetch, the contents of the Program Counter (PC) are transferred to: Medium

(A) Instruction Register (IR)
(B) Memory Address Register (MAR)
(C) Accumulator (AC)
(D) Data Register (DR)

Correct Answer: [B]
Explanation: Fetch cycle ka pehla step: MAR <- PC (PC ka address MAR mein load hota hai taaki memory se instruction read kiya ja sake).

Q87. Which of the following generates the master timing signals in a CPU? Easy

(A) Program Counter
(B) Clock Generator
(C) Instruction Decoder
(D) Memory Buffer

Correct Answer: [B]
Explanation: System Clock continuous periodic pulses generate karta hai jo registers aur flip-flops ke operations ko synchronize karta hai.

UNIT 7: BOOLEAN ALGEBRA - PRACTICE MCQS

Q88. According to the Idempotent Law of Boolean algebra, A + A is equal to: Easy

(A) 1
(B) 0
(C) A
(D) 2A

Correct Answer: [C]
Explanation: Idempotent law states: A + A = A and A * A = A.

Q89. The Boolean expression A + A'B simplifies to: Medium

(A) A
(B) A + B
(C) AB
(D) B

Correct Answer: [B]
Explanation: Absorption / Distributive law: A + A'B = (A + A')(A + B) = 1 * (A + B) = A + B.

Q90. The Dual of the Boolean expression (A + B) * (C + D) is: Medium

(A) (A * B) + (C * D)
(B) (A' + B') * (C' + D')
(C) (A * B) * (C * D)
(D) AB + CD

Correct Answer: [A]
Explanation: Duality rule: Replace AND with OR, and OR with AND. Variables remain uncomplemented! (A + B) * (C + D) becomes (A * B) + (C * D).

Q91. The simplified form of the Boolean expression AB + AB' is: Easy

(A) A
(B) B
(C) A + B
(D) 1

Correct Answer: [A]
Explanation: AB + AB' = A(B + B') = A(1) = A.

Q92. De Morgan's theorem states that (A + B)' is equal to: Easy

(A) A' + B'
(B) A' * B'
(C) (A * B)'
(D) A * B

Correct Answer: [B]
Explanation: (A + B)' = A' * B' (The complement of a sum equals the product of complements).

Q93. De Morgan's theorem states that (A * B)' is equal to: Easy

(A) A' * B'
(B) A' + B'
(C) A + B
(D) (A + B)'

Correct Answer: [B]
Explanation: (A * B)' = A' + B' (The complement of a product equals the sum of complements).

Q94. According to the Consensus Theorem, AB + A'C + BC is equal to: Medium

(A) AB + BC
(B) A'C + BC
(C) AB + A'C
(D) A + B + C

Correct Answer: [C]
Explanation: Consensus theorem: AB + A'C + BC = AB + A'C. The term BC is redundant and eliminated.

Q95. The Boolean expression P + QR is the reduced form of which expression? Medium

(A) (P + Q)R
(B) (P + R)Q
(C) (P + Q)(P + R)
(D) PQ + QR

Correct Answer: [C]
Explanation: Distributive law: (P + Q)(P + R) = P * P + PR + PQ + QR = P + PR + PQ + QR = P(1 + R + Q) + QR = P + QR.

Q96. Simplify the expression: Y = A'BC + A'BC' + AB'C + ABC. Hard

(A) A'B + AC
(B) B + C
(C) A'B + C
(D) A + BC

Correct Answer: [A]
Explanation: A'B(C + C') + AC(B' + B) = A'B(1) + AC(1) = A'B + AC.

Q97. The value of Boolean expression A * (A + B) is: Easy

(A) A
(B) B
(C) AB
(D) A + B

Correct Answer: [A]
Explanation: Absorption law: A * (A + B) = A * A + AB = A + AB = A(1 + B) = A.

Q98. According to Boolean algebra, A + 1 is equal to: Easy

(A) A
(B) 1
(C) 0
(D) A'

Correct Answer: [A]
Explanation: Wait! A + 1 is 1! (Null / Domination law: 0+1=1, 1+1=1, so A+1 = 1).

Q99. According to Boolean algebra, A * 0 is equal to: Easy

(A) 0
(B) A
(C) 1
(D) A'

Correct Answer: [A]
Explanation: Null / Domination law for AND: Anything ANDed with 0 is always 0.

Q100. The Involution Law in Boolean algebra states: Easy

(A) A + A = A
(B) (A')' = A
(C) A * A' = 0
(D) A + 0 = A

Correct Answer: [B]
Explanation: Involution law double complementation ko state karta hai: (A')' = A.

Q101. The complement of the expression F = AB + C'D is: Medium

(A) (A' + B')(C + D')
(B) (A' + B') + (C + D')
(C) A'B' + CD'
(D) (AB)'(C'D)'

Correct Answer: [A]
Explanation: F' = (AB + C'D)' = (AB)' * (C'D)' = (A' + B') * ((C')' + D') = (A' + B')(C + D').

Q102. Simplify: F = (A + B)(A + B'): Easy

(A) A
(B) B
(C) AB
(D) 0

Correct Answer: [A]
Explanation: (A + B)(A + B') = A + BB' = A + 0 = A.

Q103. What is the dual of the identity A + A' = 1? Easy

(A) A * A' = 0
(B) A + A = 1
(C) A * A = 0
(D) A' + A = 1

Correct Answer: [A]
Explanation: Replace + with * and 1 with 0: A * A' = 0.

Q104. Simplify the expression: X = A'B'C + A'BC + AB'C + ABC. Medium

(A) A
(B) B
(C) C
(D) A + B

Correct Answer: [C]
Explanation: Factor C common: C(A'B' + A'B + AB' + AB) = C(A'(B' + B) + A(B' + B)) = C(A'(1) + A(1)) = C(A' + A) = C(1) = C.

Q105. The Boolean expression (X + Y)(X' + Z)(Y + Z) reduces to: Hard

(A) (X + Y)(X' + Z)
(B) (X + Y)(Y + Z)
(C) (X' + Z)(Y + Z)
(D) XYZ

Correct Answer: [A]
Explanation: Dual of Consensus theorem: (X + Y)(X' + Z)(Y + Z) = (X + Y)(X' + Z). Term (Y + Z) is redundant.

Q106. In Boolean algebra, what is the value of 1 + 1? Easy

(A) 1
(B) 2
(C) 0
(D) 10

Correct Answer: [A]
Explanation: In Boolean logic, 1 + 1 represents Logic 1 OR Logic 1 = 1 (NOT arithmetic 2).

Q107. Simplify: Y = A + AB + AB'C: Easy

(A) A
(B) B
(C) C
(D) AB

Correct Answer: [A]
Explanation: Y = A(1 + B + B'C). In Boolean algebra, 1 + anything = 1. So Y = A(1) = A.

UNIT 8: LOGIC GATES - PRACTICE MCQS

Q108. Which of the following gates are called UNIVERSAL GATES? Easy

(A) AND and OR
(B) NAND and NOR
(C) XOR and XNOR
(D) NOT and AND

Correct Answer: [B]
Explanation: NAND aur NOR universal gates hain kyunki kisi bhi Boolean function ko sirf aur sirf inhi gates se implement kiya ja sakta hai.

Q109. What is the minimum number of 2-input NAND gates required to implement a 2-input XOR gate? Medium

(A) 3
(B) 4
(C) 5
(D) 6

Correct Answer: [B]
Explanation: 2-input XOR gate implement karne ke liye exactly 4 NAND gates lagte hain.

Q110. What is the minimum number of 2-input NOR gates required to implement a 2-input XOR gate? Medium

(A) 4
(B) 5
(C) 6
(D) 3

Correct Answer: [B]
Explanation: XOR gate using NOR requires 5 NOR gates! (While XOR using NAND requires 4).

Q111. What is the minimum number of 2-input NAND gates required to implement a 2-input XNOR gate? Medium

(A) 4
(B) 5
(C) 6
(D) 3

Correct Answer: [B]
Explanation: XNOR using NAND requires 5 NAND gates! (XNOR using NOR requires 4 NOR gates).

Q112. A logic circuit produces output 1 ONLY when all its inputs are 0. The gate is a/an: Easy

(A) AND Gate
(B) NAND Gate
(C) NOR Gate
(D) XOR Gate

Correct Answer: [C]
Explanation: NOR gate output is 1 only when all inputs are 0 (Y = (0 + 0)' = 1).

Q113. An XOR gate acts as an odd-parity detector because its output is 1 when: Easy

(A) An even number of inputs are 1
(B) An odd number of inputs are 1
(C) All inputs are 1
(D) All inputs are 0

Correct Answer: [B]
Explanation: XOR gate is an ODD function: multiple inputs mein jab total 1s odd hote hain to output 1 hota hai.

Q114. Given the truth table: A=0, B=0 -> X=1; A=0, B=1 -> X=0; A=1, B=0 -> X=0; A=1, B=1 -> X=0. Which Boolean function does it represent? Easy

(A) OR
(B) XOR
(C) NOR
(D) XNOR

Correct Answer: [C]
Explanation: Output 1 sirf 0, 0 par hai. This is NOR gate: (A + B)'.

Q115. If one input of a 2-input XOR gate is connected to Logic 1, the output is: Medium

(A) 0
(B) 1
(C) A
(D) A' (Inverter)

Correct Answer: [D]
Explanation: A XOR 1 = A * 1' + A' * 1 = A * 0 + A' * 1 = A'. Thus it acts as an Inverter (NOT gate).

Q116. If one input of a 2-input XOR gate is connected to Logic 0, the output is: Easy

(A) 0
(B) 1
(C) A (Buffer)
(D) A'

Correct Answer: [C]
Explanation: A XOR 0 = A * 0' + A' * 0 = A * 1 + 0 = A. Thus it acts as a Buffer.

Q117. How many 2-input NAND gates are needed to implement an AND gate? Easy

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: [B]
Explanation: First NAND produces (AB)'. Second NAND as inverter produces ((AB)')' = AB. Total = 2 NAND gates.

Q118. How many 2-input NAND gates are needed to implement an OR gate? Easy

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: [B]
Explanation: A' and B' are generated using 2 NAND gates. A third NAND gate takes (A' * B')' = A + B. Total = 3 NAND gates.

Q119. The Boolean expression for a 2-input XNOR gate is: Easy

(A) AB + A'B'
(B) A'B + AB'
(C) A'B' + AB
(D) Both A and C

Correct Answer: [D]
Explanation: XNOR = (A XOR B)' = AB + A'B' = A'B' + AB. Options A and C are identical.

Q120. In positive logic convention: Easy

(A) Higher voltage represents Logic 1, lower voltage represents Logic 0
(B) Higher voltage represents Logic 0, lower voltage represents Logic 1
(C) Negative voltages represent Logic 1
(D) Logic values are inverted

Correct Answer: [A]
Explanation: Positive logic mein high voltage level = Logic 1, low voltage level = Logic 0.

Q121. A positive logic AND gate becomes what gate in negative logic? Hard

(A) Negative logic AND gate
(B) Negative logic OR gate
(C) Negative logic NAND gate
(D) Negative logic XOR gate

Correct Answer: [B]
Explanation: By De Morgan's duality, positive logic AND gate corresponds to negative logic OR gate.

Q122. What is the output Y of an XNOR gate when inputs A = 1 and B = 1? Easy

(A) 0
(B) 1
(C) High Impedance
(D) Undefined

Correct Answer: [B]
Explanation: XNOR is equivalence detector: when inputs are identical (1,1 or 0,0), output is 1.

Q123. A gate that produces Logic 0 only when all its inputs are Logic 1 is a/an: Easy

(A) AND Gate
(B) NAND Gate
(C) NOR Gate
(D) XOR Gate

Correct Answer: [B]
Explanation: NAND gate output is 0 only when all inputs are 1; otherwise it is 1.

Q124. The propagation delay of a logic gate is defined as: Medium

(A) Time taken for input to rise from 0 to 1
(B) Time delay between transition of input and corresponding transition of output
(C) Time the gate remains powered off
(D) Clock cycle time

Correct Answer: [B]
Explanation: Propagation delay (t_pd) input change hone aur uske response mein output change hone ke beech ka physical transition delay hota hai.

Q125. Which gate is also known as an "Equivalence Gate"? Easy

(A) XOR Gate
(B) XNOR Gate
(C) NAND Gate
(D) NOR Gate

Correct Answer: [B]
Explanation: XNOR gate ko Equivalence gate ya Coincidence gate kehte hain kyunki ye dono inputs equal hone par 1 deta hai.

Q126. Simplify the logic circuit expression: Y = ((A * B)' * (A * B)')': Medium

(A) A * B
(B) (A * B)'
(C) A + B
(D) 0

Correct Answer: [A]
Explanation: Let X = (A * B)'. Expression is (X * X)' = (X)' = ((A * B)')' = A * B.

Q127. How many 2-input NOR gates are required to implement a NOT gate (Inverter)? Easy

(A) 1
(B) 2
(C) 3
(D) 4

Correct Answer: [A]
Explanation: NOR gate ke dono inputs ko short kar do: (A + A)' = A'. Exactly 1 NOR gate!

UNIT 9: CANONICAL FORMS - PRACTICE MCQS

Q128. For an n-variable Boolean function, the total number of distinct minterms is: Easy

(A) n
(B) 2n
(C) 2^n
(D) n^2

Correct Answer: [C]
Explanation: n variables se total 2^n possible input combinations bante hain, isliye total minterms = 2^n.

Q129. In standard SOP form, the minterm corresponding to binary input combination (1, 0, 1) for variables (A, B, C) is: Easy

(A) A B' C
(B) A' B C'
(C) A + B' + C
(D) A' + B + C'

Correct Answer: [A]
Explanation: Minterm SOP rule: 1 = uncomplemented (A, C), 0 = complemented (B'). So m_5 = A B' C.

Q130. In standard POS form, the maxterm corresponding to binary input (1, 0, 1) for variables (A, B, C) is: Medium

(A) A B' C
(B) A' + B + C'
(C) A + B' + C
(D) A' B C'

Correct Answer: [B]
Explanation: Maxterm POS rule: 0 = normal variable, 1 = complemented variable! For 1, 0, 1: M_5 = A' + B + C'.

Q131. If a 3-variable function is given as F(A, B, C) = Sigma m(0, 2, 4, 6), its canonical POS representation is: Medium

(A) Pi M(1, 3, 5, 7)
(B) Pi M(0, 2, 4, 6)
(C) Sigma m(1, 3, 5, 7)
(D) Pi M(0, 1, 2, 3)

Correct Answer: [A]
Explanation: A 3-variable function has terms 0 to 7. Bachi hui missing terms POS form banati hain: Pi M(1, 3, 5, 7).

Q132. The relationship between minterm m_i and maxterm M_i of the same index is: Easy

(A) M_i = m_i
(B) M_i = (m_i)' (Complements of each other)
(C) M_i + m_i = 0
(D) M_i * m_i = 1

Correct Answer: [B]
Explanation: By De Morgan's laws, maxterm M_i is the exact complement of minterm m_i: M_i = (m_i)'.

Q133. How many maxterms does a 4-variable Boolean function have in total? Easy

(A) 4
(B) 8
(C) 16
(D) 32

Correct Answer: [C]
Explanation: For n = 4 variables, total maxterms = 2^4 = 16 (M_0 to M_15).

Q134. Convert the non-canonical SOP term AB into canonical minterms for a 3-variable function F(A, B, C): Medium

(A) ABC + ABC'
(B) ABC + AB'C
(C) A'BC + ABC
(D) AB + ABC

Correct Answer: [A]
Explanation: Multiply missing variable term (C + C'): AB(C + C') = ABC + ABC' (which are m_7 and m_6).

Q135. A Boolean function F is self-dual if: Medium

(A) F = F'
(B) F is equal to its dual F_D
(C) F = 0
(D) F = 1

Correct Answer: [B]
Explanation: Self-dual function ki property hoti hai ki function ka dual lene par wahi same function mil jata hai (F_D = F).

Q136. How many self-dual Boolean functions of n variables can be formed? Hard

(A) 2^n
(B) 2^(2^(n-1))
(C) 2^(2^n)
(D) 2^(n-1)

Correct Answer: [B]
Explanation: Standard formula for number of self-dual functions of n variables: 2^(2^(n-1)). For n=3: 2^(2^2) = 2^4 = 16.

Q137. The complement of function F = Sigma m(1, 3, 5) for 3 variables is: Medium

(A) Sigma m(0, 2, 4, 6, 7)
(B) Pi M(1, 3, 5)
(C) Both A and B
(D) Sigma m(1, 3, 5)

Correct Answer: [C]
Explanation: Complement F' contains all terms where F=0: F' = Sigma m(0, 2, 4, 6, 7) = Pi M(1, 3, 5). Both are correct.

Q138. Minterm m_0 for 3 variables (x, y, z) is written as: Easy

(A) x' y' z'
(B) x + y + z
(C) x y z
(D) x' + y' + z'

Correct Answer: [A]
Explanation: m_0 corresponds to 0, 0, 0: x' y' z'.

UNIT 10 & 11: K-MAPS - PRACTICE MCQS

Q139. Why is Gray code ordering used to label the rows and columns of a K-Map? Easy

(A) It is easier to write
(B) Only one bit changes between adjacent cells to enable minimization
(C) It sorts cells in decimal ascending order
(D) It uses fewer transistors

Correct Answer: [B]
Explanation: Gray code ensure karta hai ki pass wali do cells mein sirf ek variable change ho, jisse A + A' = 1 factor out ho sake.

Q140. In a 4-variable K-Map, grouping 8 adjacent cells (an Octet) eliminates how many variables? Medium

(A) 1 variable
(B) 2 variables
(C) 3 variables
(D) 4 variables

Correct Answer: [C]
Explanation: A group of 2^k cells eliminates k variables. For an octet (8 = 2^3 cells), k = 3 variables are eliminated.

Q141. In a 4-variable K-Map, grouping 4 adjacent cells (a Quad) eliminates how many variables? Easy

(A) 1 variable
(B) 2 variables
(C) 3 variables
(D) 4 variables

Correct Answer: [B]
Explanation: A group of 4 cells (2^2) eliminates 2 variables, leaving a 2-variable product term.

Q142. What is a Prime Implicant (PI) in K-Map simplification? Medium

(A) Any single minterm
(B) The largest possible group of 2^k cells that cannot be combined into a larger group
(C) Only groups of 16 cells
(D) A group that contains only don't cares

Correct Answer: [B]
Explanation: Prime Implicant aisa maximal rectangle of 1s hota hai jise kisi aur bade group mein expand na kiya ja sake.

Q143. An Essential Prime Implicant (EPI) is defined as a prime implicant that: Medium

(A) Has the maximum number of cells
(B) Contains at least one '1' that is not covered by any other prime implicant
(C) Contains no don't-care conditions
(D) Is diagonal

Correct Answer: [B]
Explanation: EPI wo prime implicant hai jisme kam se kam ek aisa 1 ho jo kisi aur group mein shamil na ho.

Q144. In a 4-variable K-Map (variables A, B, C, D), which four corner cells can be combined into a single Quad? Medium

(A) 0, 1, 2, 3
(B) 0, 4, 8, 12
(C) 0, 2, 8, 10
(D) 0, 3, 12, 15

Correct Answer: [C]
Explanation: The 4 corner cells are m_0 (0000), m_2 (0010), m_8 (1000), and m_10 (1010). They form a valid corner quad giving term B' D'.

Q145. When simplifying a Boolean function with Don't Care conditions using K-Map, we should: Medium

(A) Always group all don't cares into groups
(B) Never include don't cares in any group
(C) Use don't cares only when they help to make a group larger
(D) Invert all don't cares to 0

Correct Answer: [C]
Explanation: Don't cares ko sirf tabhi 1 treat karte hain jab unse group ka size bada ho raha ho; bache hue don't cares ko ignore kar dete hain.

Q146. How many cells are required in a Karnaugh map for a function with 5 variables? Easy

(A) 16 cells
(B) 32 cells
(C) 64 cells
(D) 25 cells

Correct Answer: [B]
Explanation: Total cells = 2^n = 2^5 = 32 cells.

Q147. Which of the following group sizes is NOT valid in K-Map grouping? Easy

(A) 2
(B) 4
(C) 6
(D) 8

Correct Answer: [C]
Explanation: Group sizes must strictly be powers of 2 (1, 2, 4, 8, 16...). A group of 6 is INVALID!

Q148. Simplify the 3-variable function: F(x, y, z) = Sigma m(0, 2, 4, 6) using K-Map: Medium

(A) z'
(B) x'
(C) y'
(D) z

Correct Answer: [A]
Explanation: Minterms 0, 2, 4, 6 are all minterms where z = 0! In K-map they form a quad across both rows in columns 00 and 10. Result = z'.

Q149. In K-Map POS minimization, groups are formed of: Easy

(A) 1s
(B) 0s
(C) Xs only
(D) Both 0s and 1s together

Correct Answer: [B]
Explanation: SOP minimization mein 1s ko group kiya jata hai; POS minimization mein 0s ko group kiya jata hai.

Q150. What is the term resulting from combining cells 0, 1, 2, 3 in a 4-variable K-Map (A, B, C, D)? Medium

(A) A'B'
(B) C'D'
(C) AB
(D) CD

Correct Answer: [A]
Explanation: Cells 0, 1, 3, 2 form the entire first row where A=0 and B=0. So the term is A' B'.

UNIT 12: ADDERS & SUBTRACTORS - PRACTICE MCQS

Q151. A logic circuit that can add two 1-bit numbers and produce sum and carry, but CANNOT handle a carry-in from a previous stage, is called a: Easy

(A) Half Adder
(B) Full Adder
(C) Parallel Adder
(D) Ripple Adder

Correct Answer: [A]
Explanation: Half Adder do inputs A aur B ko add karta hai par previous stage ka carry input nahi le sakta. Full adder carry-in bhi leta hai.

Q152. What are the correct Boolean expressions for the Sum and Carry outputs of a Half Adder? Easy

(A) Sum = A * B, Carry = A + B
(B) Sum = A XOR B, Carry = A * B
(C) Sum = A + B, Carry = A XOR B
(D) Sum = (A XOR B)', Carry = A * B

Correct Answer: [B]
Explanation: Half Adder: Sum = A XOR B = A'B + AB', Carry = A * B.

Q153. How many Half Adders and OR gates are required to construct a single Full Adder? Easy

(A) 1 Half Adder and 1 OR Gate
(B) 2 Half Adders and 1 OR Gate
(C) 2 Half Adders and 2 OR Gates
(D) 3 Half Adders and 1 OR Gate

Correct Answer: [B]
Explanation: Standard hardware construction: 1 Full Adder = 2 Half Adders + 1 OR Gate.

Q154. The Boolean expression for the Carry Out (C_out) of a Full Adder with inputs A, B, and C_in is: Easy

(A) AB + BC_in + AC_in
(B) A XOR B XOR C_in
(C) A'B + BC_in
(D) AB(A + B + C_in)

Correct Answer: [A]
Explanation: Full Adder Carry equation: C_out = AB + BC_in + AC_in = AB + C_in(A XOR B).

Q155. What is the Sum (S) output of a Half Adder when both inputs are 1 (A = 1, B = 1)? Easy

(A) 0
(B) 1
(C) 2
(D) Undefined

Correct Answer: [A]
Explanation: Sum = A XOR B = 1 XOR 1 = 0 (Carry = 1 * 1 = 1).

Q156. The Difference (D) and Borrow (B) expressions for a Half Subtractor (A - B) are: Easy

(A) D = A XOR B, B = A' * B
(B) D = A * B, B = A XOR B
(C) D = A XOR B, B = A * B'
(D) D = A + B, B = A'B

Correct Answer: [A]
Explanation: Half Subtractor (A - B): Difference D = A XOR B, Borrow B = A' * B.

Q157. To add two 4-bit binary numbers, how many Full Adders are required in a ripple carry adder? Medium

(A) 2 Full Adders
(B) 3 Full Adders and 1 Half Adder
(C) 4 Full Adders (or 3 FA + 1 HA)
(D) 4 Half Adders

Correct Answer: [C]
Explanation: Either 4 Full Adders (if Cin is needed for LSB), or 1 Half Adder (for LSB) + 3 Full Adders.

Q158. The main speed limitation of a Parallel Ripple Carry Adder is due to: Medium

(A) Clock jitter
(B) Propagation delay of carry rippling through each full adder stage
(C) Excessive fan-out
(D) Power dissipation

Correct Answer: [B]
Explanation: Ripple carry adder mein har stage ko previous stage ke carry ka wait karna padta hai, jisse total delay = n * t_carry ho jata hai.

Q159. Which circuit overcomes the carry propagation delay of a ripple carry adder? Easy

(A) Serial Adder
(B) Carry Lookahead Adder (CLA)
(C) Half Subtractor
(D) Ring Counter

Correct Answer: [B]
Explanation: Carry Lookahead Adder (CLA) carry generate (G_i = A_i * B_i) aur carry propagate (P_i = A_i XOR B_i) terms calculate karke saare carries parallel generate karta hai.

Q160. What is the minimum number of 2-input NAND gates required to build a Half Adder? Medium

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: [B]
Explanation: Half Adder ko universal NAND gates se implement karne ke liye minimum 5 NAND gates lagte hain (4 for XOR, 1 for AND using internal tap).

Q161. How many Half Subtractors and OR gates are needed to construct a Full Subtractor? Easy

(A) 2 Half Subtractors and 1 OR Gate
(B) 1 Half Subtractor and 2 OR Gates
(C) 2 Half Subtractors and 2 OR Gates
(D) 3 Half Subtractors and 1 OR Gate

Correct Answer: [A]
Explanation: 1 Full Subtractor = 2 Half Subtractors + 1 OR Gate (Analogous to Full Adder).

Q162. The Difference output of a Full Subtractor with inputs A, B, and Borrow-in B_in is: Easy

(A) A XOR B XOR B_in
(B) A'B + AB_in
(C) A + B + B_in
(D) (A XOR B)' * B_in

Correct Answer: [A]
Explanation: Full Subtractor difference equation is identical to Full Adder sum equation: D = A XOR B XOR B_in.

Q163. Which of the following circuits is purely COMBINATIONAL? Easy

(A) Flip-Flop
(B) Shift Register
(C) Full Adder
(D) Modulo-8 Counter

Correct Answer: [C]
Explanation: Full adder combinational circuit hai (no memory, no clock). Flip-flops, shift registers, aur counters sequential circuits hain.

Q164. In a Carry Lookahead Adder, the Carry Generate term G_i is defined as: Medium

(A) A_i + B_i
(B) A_i * B_i
(C) A_i XOR B_i
(D) A_i' * B_i

Correct Answer: [B]
Explanation: G_i = A_i * B_i (carry hamesha generate hoga jab dono inputs 1 hon).

Q165. In a Carry Lookahead Adder, the Carry Propagate term P_i is defined as: Medium

(A) A_i XOR B_i (or A_i + B_i)
(B) A_i * B_i
(C) A_i' * B_i'
(D) A_i + B_i'

Correct Answer: [A]
Explanation: P_i = A_i XOR B_i (agar carry-in aata hai to wo aage propagate ho jayega).

UNIT 13: MUX, DEMUX & DECODERS - PRACTICE MCQS

Q166. How many select lines are required for a 16-to-1 Multiplexer? Easy

(A) 2
(B) 3
(C) 4
(D) 5

Correct Answer: [C]
Explanation: Formula: 2^n inputs require n select lines. Since 2^4 = 16, exactly 4 select lines are needed.

Q167. A Multiplexer is also commonly known as a: Easy

(A) Data Distributor
(B) Data Selector
(C) Code Converter
(D) Bistable Multivibrator

Correct Answer: [B]
Explanation: MUX ko Data Selector kehte hain kyunki ye kai inputs mein se ek ko select karke output par forward karta hai.

Q168. How many 2-to-1 Multiplexers are needed to construct an 8-to-1 Multiplexer? Medium

(A) 4
(B) 6
(C) 7
(D) 8

Correct Answer: [C]
Explanation: Tree calculation: First stage = 8/2 = 4 MUX; Second stage = 4/2 = 2 MUX; Final stage = 2/2 = 1 MUX. Total = 4 + 2 + 1 = 7 MUX.

Q169. A Demultiplexer with 3 select lines will have how many output lines? Easy

(A) 3
(B) 6
(C) 8
(D) 16

Correct Answer: [C]
Explanation: Output lines = 2^n = 2^3 = 8 output lines.

Q170. A 3-to-8 Decoder has how many input lines and output lines respectively? Easy

(A) 3 inputs, 8 outputs
(B) 8 inputs, 3 outputs
(C) 3 inputs, 3 outputs
(D) 8 inputs, 8 outputs

Correct Answer: [A]
Explanation: n-to-2^n decoder: 3 inputs decode into 2^3 = 8 outputs.

Q171. Which combinational circuit can be used directly as a Demultiplexer if its Enable input is used as the data line? Medium

(A) Encoder
(B) Decoder
(C) Multiplexer
(D) Priority Encoder

Correct Answer: [B]
Explanation: A Decoder with an Enable pin behaves exactly as a Demultiplexer when data is fed to Enable.

Q172. Which circuit converts 2^n input lines into n binary output lines? Easy

(A) Decoder
(B) Encoder
(C) Multiplexer
(D) Demultiplexer

Correct Answer: [B]
Explanation: Encoder 2^n active inputs ko n-bit binary code mein encode karta hai.

Q173. In a Priority Encoder, if multiple inputs are active simultaneously, the output corresponds to: Easy

(A) The lowest priority input
(B) The highest priority input
(C) An error code
(D) A random combination

Correct Answer: [B]
Explanation: Priority encoder resolves ambiguity by giving precedence to the highest priority active input.

Q174. How many 4-to-1 Multiplexers are required to construct a 16-to-1 Multiplexer? Medium

(A) 4
(B) 5
(C) 6
(D) 7

Correct Answer: [B]
Explanation: First stage = 16/4 = 4 MUX; Second stage = 4/4 = 1 MUX. Total = 4 + 1 = 5 MUX.

Q175. Match Column A with Column B: Easy
(1) Multiplexer -> (a) Selects 1 from many
(2) Demultiplexer -> (b) 1 to many distributor
(3) Decoder -> (c) n to 2^n outputs
(4) Encoder -> (d) 2^n to n inputs

(A) 1-a, 2-b, 3-c, 4-d
(B) 1-b, 2-a, 3-d, 4-c
(C) 1-a, 2-c, 3-b, 4-d
(D) 1-d, 2-c, 3-b, 4-a

Correct Answer: [A]
Explanation: MUX = 1 from many; DEMUX = 1 to many; Decoder = n to 2^n; Encoder = 2^n to n.

Q176. Can any Boolean function of n variables be implemented using a 2^(n-1)-to-1 MUX? Hard

(A) Yes, by using (n-1) variables as select lines and 1 variable as data inputs
(B) No, at least 2^n MUX is required
(C) Only if the function is self-dual
(D) Only for symmetric functions

Correct Answer: [A]
Explanation: Yes! Using Shannon's expansion, n variables function can be implemented using 2^(n-1)-to-1 MUX by feeding 0, 1, variable or variable complement to inputs.

Q177. How many 3-to-8 Decoders are needed to build a 4-to-16 Decoder? Medium

(A) 1
(B) 2 with 1 inverter
(C) 3
(D) 4

Correct Answer: [B]
Explanation: Two 3-to-8 decoders enabled by the 4th input bit (one directly, one through NOT gate). Total = 2 decoders + 1 inverter.

Q178. An Octal-to-Binary encoder has how many input and output lines? Easy

(A) 8 inputs, 3 outputs
(B) 3 inputs, 8 outputs
(C) 8 inputs, 4 outputs
(D) 16 inputs, 4 outputs

Correct Answer: [A]
Explanation: Octal has 8 symbols (0 to 7), so 8 input lines and 3 binary output lines (2^3 = 8).

Q179. A Decimal-to-BCD Priority Encoder has how many input lines? Easy

(A) 4 inputs
(B) 8 inputs
(C) 10 inputs
(D) 16 inputs

Correct Answer: [C]
Explanation: Decimal digits are 0 to 9, so 10 input lines and 4 BCD output lines.

Q180. Which circuit is widely used for parallel-to-serial data conversion? Medium

(A) Decoder
(B) Encoder
(C) Multiplexer
(D) Demultiplexer

Correct Answer: [C]
Explanation: Multiplexer parallel data bits ko select lines ke sequential scanning ke zariye single serial output line par bhejta hai.

UNIT 14: SEQUENTIAL CIRCUITS & LATCHES - PRACTICE MCQS

Q181. The output of a sequential circuit depends on: Easy

(A) Present inputs only
(B) Past outputs (present state) only
(C) Both present inputs and past outputs (memory)
(D) External temperature

Correct Answer: [C]
Explanation: Sequential circuits memory elements rakhte hain, isliye output present inputs aur previous stored state dono par depend karta hai.

Q182. The primary difference between a Latch and a Flip-Flop is: Easy

(A) Latch is edge-triggered, Flip-flop is level-triggered
(B) Latch is level-triggered, Flip-flop is edge-triggered
(C) Latch has memory, Flip-flop does not
(D) Flip-flop is purely analog

Correct Answer: [B]
Explanation: Latch level-triggered hota hai (active during clock level); Flip-Flop edge-triggered hota hai (active only at clock transitions).

Q183. In an SR latch constructed using NOR gates, what happens when both S = 1 and R = 1? Medium

(A) No change (Hold state)
(B) Reset (Q = 0)
(C) Set (Q = 1)
(D) Invalid / Forbidden condition (both outputs become 0)

Correct Answer: [D]
Explanation: NOR latch mein S=1, R=1 forbidden state hoti hai kyunki dono outputs Q aur Q' simultaneously 0 ban jate hain.

Q184. In an SR latch constructed using NAND gates (active-low inputs S' and R'), the forbidden state occurs when: Medium

(A) S' = 0, R' = 0
(B) S' = 0, R' = 1
(C) S' = 1, R' = 0
(D) S' = 1, R' = 1

Correct Answer: [A]
Explanation: NAND latch active-low hota hai: S'=0, R'=0 par forbidden state aati hai (both outputs forced to 1).

Q185. Positive edge-triggering refers to state change occurring on: Easy

(A) High clock level
(B) Falling edge (1 to 0)
(C) Rising edge (0 to 1)
(D) Low clock level

Correct Answer: [C]
Explanation: Positive edge = rising edge = transition from 0 to 1 (Low to High).

Q186. Negative edge-triggering refers to state change occurring on: Easy

(A) Rising edge (0 to 1)
(B) Falling edge (1 to 0)
(C) Constant high level
(D) Constant low level

Correct Answer: [B]
Explanation: Negative edge = falling edge = transition from 1 to 0 (High to Low).

Q187. The minimum time for which the input signal must remain stable BEFORE the active clock edge arrives is called: Medium

(A) Hold time
(B) Propagation delay
(C) Setup time
(D) Clock skew

Correct Answer: [C]
Explanation: Setup time (t_setup) wo minimum time hai jo clock edge aane se PEHLE input ko stable rehna padta hai.

Q188. The minimum time for which the input signal must remain stable AFTER the active clock edge is called: Medium

(A) Setup time
(B) Hold time
(C) Rise time
(D) Fall time

Correct Answer: [B]
Explanation: Hold time (t_hold) wo minimum time hai jo clock edge trigger hone ke BAAD input ko stable maintain rakhna padta hai.

Q189. In an asynchronous sequential circuit: Medium

(A) All operations are synchronized to a master clock
(B) State changes can occur at any time inputs change, without a common clock
(C) Propagation delay is zero
(D) Memory elements are absent

Correct Answer: [B]
Explanation: Asynchronous sequential circuits common master clock par rely nahi karte; inputs change hote hi state transitions trigger hote hain.

Q190. How many stable states does a basic bistable multivibrator have? Easy

(A) 1
(B) 2
(C) 3
(D) 0

Correct Answer: [B]
Explanation: Bistable multivibrator (Latch / Flip-Flop) ke paas exactly 2 stable states (0 and 1) hoti hain.

Q191. In a NOR-gate SR latch, what is the output state when S = 0 and R = 0? Easy

(A) Q = 1, Q' = 0
(B) Q = 0, Q' = 1
(C) No change / Hold state
(D) Undefined

Correct Answer: [C]
Explanation: NOR latch mein S=0, R=0 memory/hold state hoti hai (no change).

Q192. What is a "Glitch" in digital logic circuits? Medium

(A) A permanent short circuit
(B) An unwanted, temporary spike or pulse caused by differing propagation delays
(C) A clock failure
(D) A software bug

Correct Answer: [B]
Explanation: Glitch ek temporary hazardous spike hoti hai jo input signals ke unequal propagation delays ke karan create hoti hai.

UNIT 15: FLIP-FLOPS - PRACTICE MCQS

Q193. What is the characteristic equation of an SR Flip-Flop? Medium

(A) Q(t+1) = S + R' Q(t)
(B) Q(t+1) = S' + R Q(t)
(C) Q(t+1) = S * R Q(t)
(D) Q(t+1) = S XOR Q(t)

Correct Answer: [A]
Explanation: SR characteristic equation is Q(t+1) = S + R' Q(t) with condition S * R = 0.

Q194. What is the characteristic equation of a JK Flip-Flop? Medium

(A) Q(t+1) = J Q(t) + K' Q'(t)
(B) Q(t+1) = J Q'(t) + K' Q(t)
(C) Q(t+1) = J' Q(t) + K Q'(t)
(D) Q(t+1) = J XOR K

Correct Answer: [B]
Explanation: JK characteristic equation is Q(t+1) = J Q'(t) + K' Q(t).

Q195. In a JK Flip-Flop, when both inputs are 1 (J = 1, K = 1), the next state is: Easy

(A) No change (Hold)
(B) Reset (0)
(C) Set (1)
(D) Toggle (Complement of present state, Q')

Correct Answer: [D]
Explanation: When J = K = 1, the JK flip-flop toggles: Q(t+1) = Q'(t).

Q196. The Race-Around condition in a JK Flip-Flop occurs when: Medium

(A) J = 0, K = 0
(B) J = 1, K = 1, flip-flop is level-triggered, and clock pulse duration > propagation delay
(C) J = 1, K = 0
(D) Flip-flop is edge-triggered

Correct Answer: [B]
Explanation: Race-around condition occurs in level-triggered JK flip-flops when J=K=1 and clock width t_p > propagation delay Delta_t.

Q197. Which flip-flop architecture completely eliminates the race-around condition of a JK flip-flop? Easy

(A) SR Flip-Flop
(B) Master-Slave JK Flip-Flop
(C) Transparent Latch
(D) Unclocked Flip-Flop

Correct Answer: [B]
Explanation: Master-Slave JK flip-flop race-around condition ko eliminate karta hai kyunki Master aur Slave opposite clock phases par work karte hain.

Q198. What is the characteristic equation of a D Flip-Flop? Easy

(A) Q(t+1) = D
(B) Q(t+1) = D'
(C) Q(t+1) = D XOR Q(t)
(D) Q(t+1) = D + Q(t)

Correct Answer: [A]
Explanation: D Flip-Flop is a Data / Delay flip-flop: Next state directly follows input D: Q(t+1) = D.

Q199. What is the characteristic equation of a T Flip-Flop? Medium

(A) Q(t+1) = T
(B) Q(t+1) = T * Q(t)
(C) Q(t+1) = T XOR Q(t)
(D) Q(t+1) = T + Q'(t)

Correct Answer: [C]
Explanation: T Flip-Flop: Q(t+1) = T XOR Q(t) = T' Q(t) + T Q'(t).

Q200. How can a JK Flip-Flop be converted into a D Flip-Flop? Medium

(A) Connect J = D and K = D
(B) Connect J = D and K = D' (via NOT gate)
(C) Connect J = 1 and K = 1
(D) Connect J = 0 and K = D

Correct Answer: [B]
Explanation: Connect J = D and K = D'. When D=1 -> J=1, K=0 (Set); when D=0 -> J=0, K=1 (Reset).

Q201. How can a JK Flip-Flop be converted into a T Flip-Flop? Easy

(A) Connect J and K together to input T (J = T, K = T)
(B) Connect J = T and K = T'
(C) Connect J = 0 and K = T
(D) Connect J = 1 and K = T

Correct Answer: [A]
Explanation: Short J and K inputs together: when T=0 -> J=K=0 (Hold); when T=1 -> J=K=1 (Toggle).

Q202. If a clock signal of frequency 10 MHz is applied to a T flip-flop with T = 1, what is the output frequency? Medium

(A) 10 MHz
(B) 20 MHz
(C) 5 MHz
(D) 2.5 MHz

Correct Answer: [C]
Explanation: When T=1, the flip-flop toggles every cycle, dividing frequency by 2: f_out = f_in / 2 = 10 MHz / 2 = 5 MHz.

Q203. What are the required inputs (J, K) to transition a JK flip-flop from present state Q(t) = 1 to next state Q(t+1) = 0? Hard

(A) J = 0, K = 1
(B) J = X, K = 1
(C) J = 1, K = X
(D) J = 0, K = X

Correct Answer: [B]
Explanation: From JK excitation table: to go from 1 to 0, K must be 1 (either Reset K=1, J=0 or Toggle K=1, J=1), so J = X (don't care) and K = 1.

Q204. What are the required inputs (S, R) to transition an SR flip-flop from state 0 to 1? Medium

(A) S = 0, R = 1
(B) S = 1, R = 0
(C) S = X, R = 0
(D) S = 0, R = X

Correct Answer: [B]
Explanation: State 0 to 1 requires SET operation, so S = 1 and R = 0.

Q205. Which flip-flop has only one input and no invalid input combination? Easy

(A) SR Flip-Flop
(B) JK Flip-Flop
(C) D Flip-Flop
(D) NOR Latch

Correct Answer: [C]
Explanation: D Flip-Flop has a single input D and produces valid deterministic output for both D=0 and D=1.

Q206. In a Master-Slave JK Flip-Flop, the Master flip-flop is active during: Medium

(A) Clock High level
(B) Clock Low level
(C) Both High and Low levels
(D) Never

Correct Answer: [A]
Explanation: Master is enabled during positive clock level (Clock = 1); Slave is enabled during negative clock level (Clock = 0 via inverter).

Q207. How many flip-flops are needed to store an 8-bit byte? Easy

(A) 1
(B) 4
(C) 8
(D) 16

Correct Answer: [A]
Explanation: Each flip-flop stores exactly 1 bit. So 8 bits require 8 flip-flops.

UNIT 16: REGISTERS & COUNTERS - PRACTICE MCQS

Q208. Which type of shift register provides the fastest data transfer capability? Easy

(A) SISO (Serial-In Serial-Out)
(B) SIPO (Serial-In Parallel-Out)
(C) PISO (Parallel-In Serial-Out)
(D) PIPO (Parallel-In Parallel-Out)

Correct Answer: [D]
Explanation: PIPO (Parallel-In Parallel-Out) requires only 1 clock pulse to load all bits and 0 clock pulses to read all bits simultaneously.

Q209. How many clock pulses are required to completely enter an n-bit binary word into a SISO shift register? Easy

(A) 1 pulse
(B) n pulses
(C) n - 1 pulses
(D) 2n pulses

Correct Answer: [B]
Explanation: SISO mein har bit ek clock pulse par enter hoti hai, isliye n bits ke liye n clock pulses lagengi.

Q210. A Ring Counter formed using 4 flip-flops has how many distinct valid states? Easy

(A) 4 states
(B) 8 states
(C) 16 states
(D) 12 states

Correct Answer: [A]
Explanation: An N-bit Ring counter has exactly N states (equal to number of flip-flops). For N = 4: 4 states.

Q211. A Johnson (Twisted Ring) Counter formed using 4 flip-flops has how many distinct states? Medium

(A) 4 states
(B) 8 states
(C) 16 states
(D) 2 states

Correct Answer: [B]
Explanation: An N-bit Johnson counter has 2N states. For N = 4: 2 * 4 = 8 distinct states.

Q212. How many unused (invalid) states are there in a 4-bit Johnson counter? Medium

(A) 4
(B) 8
(C) 12
(D) 16

Correct Answer: [B]
Explanation: Total 4-bit states = 2^4 = 16. Valid states = 2N = 8. Unused states = 16 - 8 = 8 states.

Q213. How many flip-flops are required to construct a Modulo-12 counter? Medium

(A) 3 flip-flops
(B) 4 flip-flops
(C) 12 flip-flops
(D) 6 flip-flops

Correct Answer: [B]
Explanation: Condition: 2^(n-1) < N <= 2^n. For N = 12: 2^3 (8) < 12 <= 2^4 (16). So n = 4 flip-flops.

Q214. In an Asynchronous (Ripple) counter, the clock input of each flip-flop (except the first) is driven by: Easy

(A) The master external clock
(B) The output of the preceding flip-flop
(C) The power supply line
(D) The reset pin

Correct Answer: [B]
Explanation: Ripple counter mein previous flip-flop ka output agle flip-flop ki clock input banta hai.

Q215. In a Synchronous counter: Easy

(A) All flip-flops are clocked simultaneously by the same clock source
(B) Each flip-flop is clocked by the previous flip-flop
(C) No clock is used
(D) Flip-flops toggle at random times

Correct Answer: [A]
Explanation: Synchronous counter mein saare flip-flops ko common external clock ek sath trigger karti hai.

Q216. A 3-bit binary asynchronous up-counter will count through states in which sequence? Easy

(A) 000 to 111 (0 to 7)
(B) 111 to 000 (7 to 0)
(C) 001 to 100
(D) Random states

Correct Answer: [A]
Explanation: 3-bit up counter 0 to 2^3 - 1 = 0 to 7 count karta hai.

Q217. A Decade Counter (BCD Counter) counts how many distinct states? Easy

(A) 8 states (0 to 7)
(B) 10 states (0 to 9)
(C) 12 states (0 to 11)
(D) 16 states (0 to 15)

Correct Answer: [B]
Explanation: Decade counter MOD-10 counter hota hai jo 0 to 9 count karta hai aur state 1010 aate hi reset ho jata hai.

UNIT 17: MEMORY UNIT & CACHE - PRACTICE MCQS

Q218. Which memory type requires PERIODIC REFRESHING cycles to retain stored data? Easy

(A) SRAM (Static RAM)
(B) DRAM (Dynamic RAM)
(C) ROM
(D) Flash Memory

Correct Answer: [B]
Explanation: DRAM cells tiny capacitors par charge store karte hain jo leak ho jata hai, isliye DRAM ko periodic refresh karna mandatory hota hai.

Q219. CPU cache memory is typically built using which technology? Easy

(A) DRAM
(B) SRAM
(C) Magnetic Disk
(D) Optical ROM

Correct Answer: [B]
Explanation: SRAM flip-flop transistors se banti hai aur super fast hoti hai, isliye Cache memory SRAM se banayi jati hai.

Q220. Which storage devices can be used as reliable BACK-UP storage devices? (CUET PG PYQ) Easy

(A) RAM and Cache
(B) HDD (Hard Disk) and Magnetic Tape
(C) ROM and RAM
(D) Cache and Registers

Correct Answer: [B]
Explanation: HDD aur Magnetic Tape non-volatile permanent high-capacity storage provide karte hain, isliye backup ke liye use hote hain.

Q221. Cache memory functions as an intermediary speed buffer between: Easy

(A) RAM and ROM
(B) CPU and Main Memory (RAM)
(C) CPU and Hard Disk
(D) Monitor and Keyboard

Correct Answer: [B]
Explanation: Cache CPU aur RAM ke access speed gap ko bridge karne ke liye unke beech intermediary buffer ka kaam karta hai.

Q222. The principle of "Locality of Reference" that justifies cache memory consists of: Medium

(A) Static and Dynamic locality
(B) Temporal and Spatial locality
(C) Direct and Associative locality
(D) Internal and External locality

Correct Answer: [B]
Explanation: Temporal Locality (recently accessed item will be accessed again soon) aur Spatial Locality (nearby memory locations will be accessed soon).

Q223. In Direct Cache Mapping, memory block k maps to cache line number: Medium

(A) k / (Number of cache lines)
(B) k MOD (Number of cache lines)
(C) k * (Number of cache lines)
(D) Any random empty line

Correct Answer: [B]
Explanation: Direct mapping formula: Line number = (Block address) MOD (Total lines in cache).

Q224. If cache access time is 2 ns, main memory access time is 50 ns, and hit ratio is 0.9 (90%), the Average Memory Access Time (AMAT) is: Hard

(A) 5 ns
(B) 7 ns
(C) 10 ns
(D) 52 ns

Correct Answer: [B]
Explanation: AMAT = Hit_time + (1 - Hit_ratio) * Miss_penalty = 2 ns + (1 - 0.9) * 50 ns = 2 + 0.1 * 50 = 2 + 5 = 7 ns!

Q225. Which ROM type can be erased using Ultraviolet (UV) light exposure? Easy

(A) Mask ROM
(B) PROM
(C) EPROM
(D) EEPROM

Correct Answer: [C]
Explanation: EPROM (Erasable Programmable ROM) ko quartz window ke through Ultraviolet (UV) light daal kar erase kiya jata hai.

STUDENT PERFORMANCE TRACKER & SCORECARD

Ye scorecard table aapko apna actual preparation level evaluate karne mein help karegi. Har unit ke total questions attempt karke apna score calculate karein:

Section / Topic	Total Qs	Correct (+4)	Incorrect (-1)	Net Score	Target %
Units 1-4: Data Units & Number Systems	62				>= 85%
Units 5-6: Arithmetic & CPU Architecture	25				>= 80%
Units 7-11: Boolean Algebra & K-Maps	63				>= 90%
Units 12-13: Combinational Logic Circuits	30				>= 85%
Units 14-16: Sequential Logic & Counters	37				>= 80%
Unit 17: Memory Hierarchy & Cache	8				>= 75%
TOTAL CUET PG DIGITAL FUNDAMENTALS	225				>= 85%

[CONCEPT] Cutoff & Benchmark Score Analysis for Top MCA Colleges (DU, BHU, JNU, NITs):
- Score > 80% (720+ Marks out of 900): Top Rank Tier (DU MCA / BHU MCA confirmed selection)
- Score 65% - 80% (585 - 720 Marks): Strong Preparation (Central Universities admission)
- Score < 65% (< 585 Marks): Revision Needed! Weak topics ko Master Study Notes se dubara revise karein.

Cutoff & Benchmark Score Analysis